# COMPUTER SCIENCE & ENTREPRENEURSHIP 9

## UNIT 3: COMPUTER NETWORKS (COMPLETE NOTES)

### Unit 3: Computer Networks (Concepts & Components)

**Q.1: What is a network and state its primary types?**

Answer: A network is a system of two or more connected computers and devices that can share data and resources. Primary types include LAN, WAN, MAN, and PAN.

**Q.2: What are the fundamental components of a network?**

Answer: The four basic components are: 1. Hardware (devices like computers and routers), 2. Software, 3. Protocols (rules), 4. Transmission Media (cables or wireless).

**Q.3: What is the Internet and what is its significance?**

Answer: The Internet is the world's largest network connecting smaller and larger networks globally. It is also called the network of networks.

**Q.4: What is the importance of a Network Diagram?**

Answer: A network diagram provides a graphical representation of network devices and their interconnections, making it easier to understand and configure the network.

**Q.5: What is the primary purpose of a computer network?**

Answer: The primary purpose of a computer network is resource sharing, communication, and ensuring coordination.

**Q.6: What is meant by Resource Sharing?**

Answer: Network allows hardware (like printers), software, or data to be shared among multiple users.

**Q.7: What is the role of networks in reducing cost and improving efficiency?**

Answer: Networks eliminate the need to purchase separate devices, lower expenses, and enhance collective performance.

**Q.8: What is meant by effective communication and connectivity?**

Answer: It refers to providing fast data transmission, email, and collaboration facilities between devices and users.

**Q.9: What is the history of the World Wide Web (WWW)?**

Answer: The World Wide Web was created by Tim Berners-Lee in 1989, revolutionizing data access and sharing on the Internet.

**Q.10: What is Data Communication?**

Answer: The exchange of information or ideas between two or more devices is called data communication.

**Q.11: How many basic components does data communication have?**

Answer: Data communication consists of five basic components: 1. Sender, 2. Receiver, 3. Message, 4. Transmission Media, 5. Protocol.

**Q.12: Explain Sender, Receiver, and Message.**

Answer: Sender is who sends data; Receiver is who receives data; and Message is the data being exchanged.

**Q.13: What are Transmission Media and Protocols?**

Answer: Transmission media is the wired or wireless path through which data travels, while protocols are sets of rules controlling communication.

**Q.14: What is a Switch and how does it work?**

Answer: A switch is a network device that connects devices within the same network. It transmits data only to the target device using MAC addresses.

**Q.15: What is a MAC Address?**

Answer: A MAC address is a unique and permanent hardware address assigned to each network interface card connected to a network.

**Q.16: What is a Router?**

Answer: A router is a network device that connects different networks together and acts as a traffic director.

**Q.17: How does a Router work?**

Answer: A router determines the best path for data packets to reach their destination (Routing) and forwards them to the correct network.

**Q.18: How does air travel system analogy work in packet switching?**

Answer: Passengers (data) are divided into groups, can take different routes to reach the destination, and reassemble at the destination.

**Q.19: What is an Access Point (AP)?**

Answer: An access point is a network device that enables wireless clients to connect to a wired network.

## Network Topologies

**Q.20: What is an Access Point? Give an example.**

Answer: An access point acts as a repeater for data flow during setup. Example: Imagine a principal's office connected via intercom to all classrooms. It receives wireless data and sends it to the wired network.

**Q.21: What is Network Topology?**

Answer: Network topology refers to the arrangement and interconnection of various elements or devices (nodes) in a network, affecting performance and reliability.

**Q.22: What is Bus Topology?**

Answer: In bus topology, all devices share a single communication line called a bus. Each device is connected to this central cable. It is easy to build, but if the central cable fails, the entire network goes down.

**Q.23: What is Star Topology? Give an example.**

Answer: In star topology, all devices are connected to a central hub or switch. The hub receives data and forwards it to the target device. Example: A classroom network where all computers are connected to a central switch.

**Q.24: What is Ring Topology?**

Answer: In ring topology, each device is connected to two other devices via a cable forming a closed loop or ring. Data travels in one direction through each device.

**Q.25: Define Mesh Topology.**

Answer: In mesh topology, every device is connected directly to every other device, providing extremely high reliability and communication standards.

**Q.26: What are Transmission Modes and write their types?**

Answer: Transmission modes refer to the direction of data transfer between devices. Three main types are Simplex, Half-Duplex, and Full-Duplex.

## OSI Model and Layers

**Q.27: What is the OSI Model and how many layers does it have?**

Answer: OSI (Open Systems Interconnection) is a network framework used to understand how different protocols work. It consists of 7 layers.

**Q.28: Define Physical Layer (Layer 1) with an example.**

Answer: The physical layer is responsible for physical connections between devices, transmitting data as bits. Example: Ethernet cables or wireless signals like Wi-Fi.

**Q.29: Define Data Link Layer (Layer 2) with an example.**

Answer: This layer ensures error-free data transfer between devices and operates on physical addresses (MAC Address). Example: Network switch and frame transmission.

**Q.30: Define Network Layer (Layer 3) with an example.**

Answer: The network layer handles data routing between different networks to reach the correct destination. Example: Router, IP Address, and GPS.

**Q.31: Define Transport Layer (Layer 4) with an example.**

Answer: This layer ensures complete and correct data delivery from source to destination and flow control. Example: TCP (Transmission Control Protocol).

**Q.32: Define Session Layer (Layer 5) with an example.**

Answer: The session layer establishes, maintains, and terminates sessions between applications. Example: Online video call or web session connection.

**Q.33: Define Presentation Layer (Layer 6) with an example.**

Answer: This layer handles data formatting, encryption, and decryption. Example: SSL/TLS Encryption for data security.

**Q.34: Define Application Layer (Layer 7) with an example.**

Answer: This top layer interacts directly with end-users and software applications. Example: Web browsers like Google Chrome and email protocols (HTTP/SMTP).

## Other Important Network Concepts

**Q.35: What is Simplex Communication? Give an example.**

Answer: Data flows in only one direction (unidirectional). Example: Computer to keyboard or PC to printer.

**Q.36: Explain Half-Duplex Communication.**

Answer: Data can flow in both directions, but only one direction at a time. Example: Walkie-talkie.

**Q.37: What is Full-Duplex Communication? Give an example.**

Answer: Data can flow in both directions simultaneously. Example: Telephone or mobile call.

**Q.38: What is Internet Protocol version 4 (IPv4)?**

Answer: It is a network protocol version that uses 32-bit addresses (providing about 4.5 billion addresses).

**Q.39: What is Internet Protocol version 6 (IPv6)?**

Answer: It is the latest IP version using 128-bit addresses, providing unlimited addresses for devices worldwide.

**Q.40: What are Network Protocols? Give key examples.**

Answer: Protocols are sets of rules governing data communication on a network. Key examples: TCP/IP, HTTP, FTP.

**Q.41: What is the function of DHCP and DNS protocols?**

Answer: DNS translates domain names into IP addresses, while DHCP automatically assigns IP addresses to devices on a network.

**Q.42: Describe geographical network types (PAN, LAN, MAN, WAN).**

Answer: PAN is personal range, LAN covers a room or building, MAN spans a city, and WAN spans the entire world.

**Q.43: What is a Personal Area Network (PAN)? Give an example.**

Answer: Very short-range network between personal devices. Example: Smartphone and Bluetooth headset connection.

**Q.44: Define Local Area Network (LAN).**

Answer: A network connecting computers within a limited geographical area like a home, office, or building is called a LAN.

**Q.45: What is a Metropolitan Area Network (MAN)?**

Answer: Larger than LAN but smaller than WAN, spanning across an entire city.

**Q.46: Explain Wide Area Network (WAN).**

Answer: The world's largest network connecting countries and continents. The Internet is the best example.

**Q.47: What is a Campus Area Network (CAN)?**

Answer: A network connecting nearby buildings of a university or industrial organization.

**Q.48: What are real-world applications of computer networks?**

Answer: E-commerce, online education (LMS), banking, and healthcare systems effectively use networks.

**Q.49: What is Network Security and state its importance?**

Answer: Network security refers to protecting data, preventing unauthorized access and attacks, and establishing confidentiality.

## Unit 3: Exercise Short Questions

**Q.1: Define Communication.**

Answer: The exchange of information, data, or ideas between two or more devices is called communication.

**Q.2: List the key components of data communication.**

Answer: Key components include: 1. Sender, 2. Receiver, 3. Message, 4. Transmission Media, 5. Protocol.

**Q.3: What is the role of routers in a computer network?**

Answer: A router connects different networks and performs routing to deliver data packets to the correct destination.

**Q.4: What are the main functions of the Network Layer in the OSI model?**

Answer: The main function is routing data packets across the network and ensuring correct delivery from source to destination.

**Q.5: What is the purpose of DHCP?**

Answer: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network settings to devices.

**Q.6: What is the purpose of Encryption in network security?**

Answer: The purpose of encryption is to secure data so it remains protected from unauthorized individuals.

**Q.7: Explain the importance of Encryption in network security.**

Answer: Encryption transforms data into an incomprehensible code so that unauthorized parties cannot read it if intercepted, preserving confidentiality.

**Q.8: Define Star Topology.**

Answer: Star topology is a network arrangement where all devices are directly connected to a central hub or switch.

**Q.9: How do Firewalls play a role in network protection?**

Answer: A firewall monitors incoming and outgoing network traffic and blocks unauthorized access, hackers, and threats to secure the network.

**Q.10: What is the difference between Full-Duplex and Half-Duplex?**

Answer: Full-duplex allows communication in both directions simultaneously, while half-duplex communicates one direction at a time.

## Additional Short Questions Unit 3

**Q.11: What is the basic importance of data communication?**

Answer: It enables accurate, timely, and secure exchange of information.

**Q.12: Why are protocols necessary?**

Answer: Protocols establish communication rules between devices on a network for smooth data transfer.

**Q.13: What is the difference between Internet and Intranet?**

Answer: The Internet is a global public network, whereas an intranet is a restricted network of an organization or private entity.

**Q.14: What is Routing?**

Answer: Routing is the process of selecting the best and fastest path for data packets across a network.

**Q.15: What are the benefits of wireless networks?**

Answer: Wireless network installation is easy and gives devices mobility freedom.

**Q.16: State the basic benefits of networks.**

Answer: Data and resource sharing, cost reduction, and effective communication are core benefits.

**Q.17: What is the difference between Hub and Switch?**

Answer: A hub broadcasts data to all ports, whereas a switch sends data only to the requested device.

**Q.18: What does the Domain Name System (DNS) do?**

Answer: It translates names (like google.com) into IP addresses.